MILITARY SPECIFICATION

TRAILER, LIFT, AIRCRAFT ENGINE ETU-92/E; TF-39 TURBOFAN ENGINE

1. SCOPE

1.1 This specification covers an aircraft engine lift trailer, designated ETU-92/E.

2. APPLICABLE DOCUMENTS

2.1 The following documents form a part of this specification to the extent specified herein. The applicable issues of the governmental documents shall be the issues in effect on date of invitation for bids or request for proposal. The applicable issues of the nongovernmental documents shall be those which are available in the files of the procuring activity and made a part of the invitation for bids or request for proposal.

SPECIFICATIONS

Federal

GGG-P-501	Plumb-bobs

Military

MIL-P-514	Plates, Identification, Instruction and Marking, Blank
MIL-F-3541	Fittings, Lubrication
MIL-T-5021	Tests; Aircraft and Missile Welding Operators' Qualification
MIL-P-5516	Packing, Preformed, Petroleum Hydraulic Fluid Resistant, 160°F
MIL-H-5606	Hydraulic Fluid, Petroleum Base, Aircraft, Missile, and Ordnance
MIL-C-6021	Casting, Classification and Inspection of
MIL-H-6875	Heat Treatment of Steels (Aircraft Practice), Process for
MIL-M-8090	Mobility, Towed Aerospace Ground Equipment, General Requirements for
MIL-A-8421	Air Transportability Requirements, General Specification for
MIL-R-83232	Reel, Static Discharge, Grounding RMU-20/E

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STANDARDS

Military

MIL-STD-12	Abbreviations for Use on Drawings, Specifications, Standards, and in Technical Documents
MIL-STD-17	Mechanical Symbols
MIL-STD-100	Engineering Drawing Practices
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-209	Slinging Eyes and Attachments for Lifting and Tying Down Heavy Military Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-808	Finishes, Protective, and Codes, for Finishing Schemes for Ground and Ground Support Equipment
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-889	Dissimilar Metals
MS3102	Connector, Receptacle, Electric, Box Mounting
MS27610	Connector, Electrical Ground
MS28774	Retainer, Packing Backup, Single Turn, Tetrafluorethylene

Air Force-Navy Aeronautical

AN3114	Receptacle, External Power, 115/200 Volts, 3 Phase
AN3430	Cable Assembly, Power, Electrical, 115/200 Volts, 3 Phase

DRAWINGS

Lockheed-Georgia Company

4BSAS00288	Interface Control - Engine Trailer to Engine
4BSAS00290	Interface Control - Engine Trailer to Air Vehicle
4BSAS00293	Interface Control - Power Package Container to Engine Trailer
4S30029	Platform, Maintenance Aircraft Engine
4S30040	Stand, Maintenance, Aircraft Power Unit

PUBLICATION

Air Force Bulletin

518	Cargo Aircraft Compartment Dimensional Data

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Fire Protection Association

Pamphlet 70 National Electrical Code

(Applications for copies should be addressed to the National Fire Protection Association, 60 Batterymarch Street, Boston, Massachusetts 02110.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Components. The trailer shall consist of the following major components:

Item or System Name	See Requirement
Chassis	3.7
Lift mechanism	3.8
Engine mounting frame	3.9
Power system	3.10.

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials

3.4.1 Fungusproof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they are to be used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

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3.4.2 Standard steel. Steels complying with ASTM or AISI standards shall be utilized in the design and fabrication of the trailer and shall be corrosion resistant or treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be encountered during storage or normal service.

3.4.2.1 Heat treatment. Heat treatment of steel materials shall be in accordance with MIL-H-6875.

3.4.2.2 Castings. Castings shall be designed and classified in accordance with MIL-C-6021.

3.4.3 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.5 Design and construction. The trailer shall be designed for TF-39 turbofan engine installation and removal operations from the C-5A aircraft. The trailer shall be designed as a towable vehicle, compatible with the dimensional and load configuration of the TF-39 (EBU) (see 6.3) turbofan engine. It shall provide a capability to lift and maneuver the engine for attachment to the aircraft wing pylon fittings and shall be capable of reverse operation. A typical configuration of the trailer and engine is shown on figure 1. Aircraft pylon cowl door clearance dimensions and engine mounting supports are shown on figures 2 and 3. The trailer shall be compatible with engine hoisting equipment in the jet engine field maintenance shops and the maintenance stand shown on Drawing 4S30040. The trailer also shall be compatible with two engine maintenance platforms conforming to Drawing 4S30029 that have been prepositioned 130 inches apart, below the wing pylon station, at the aircraft site.

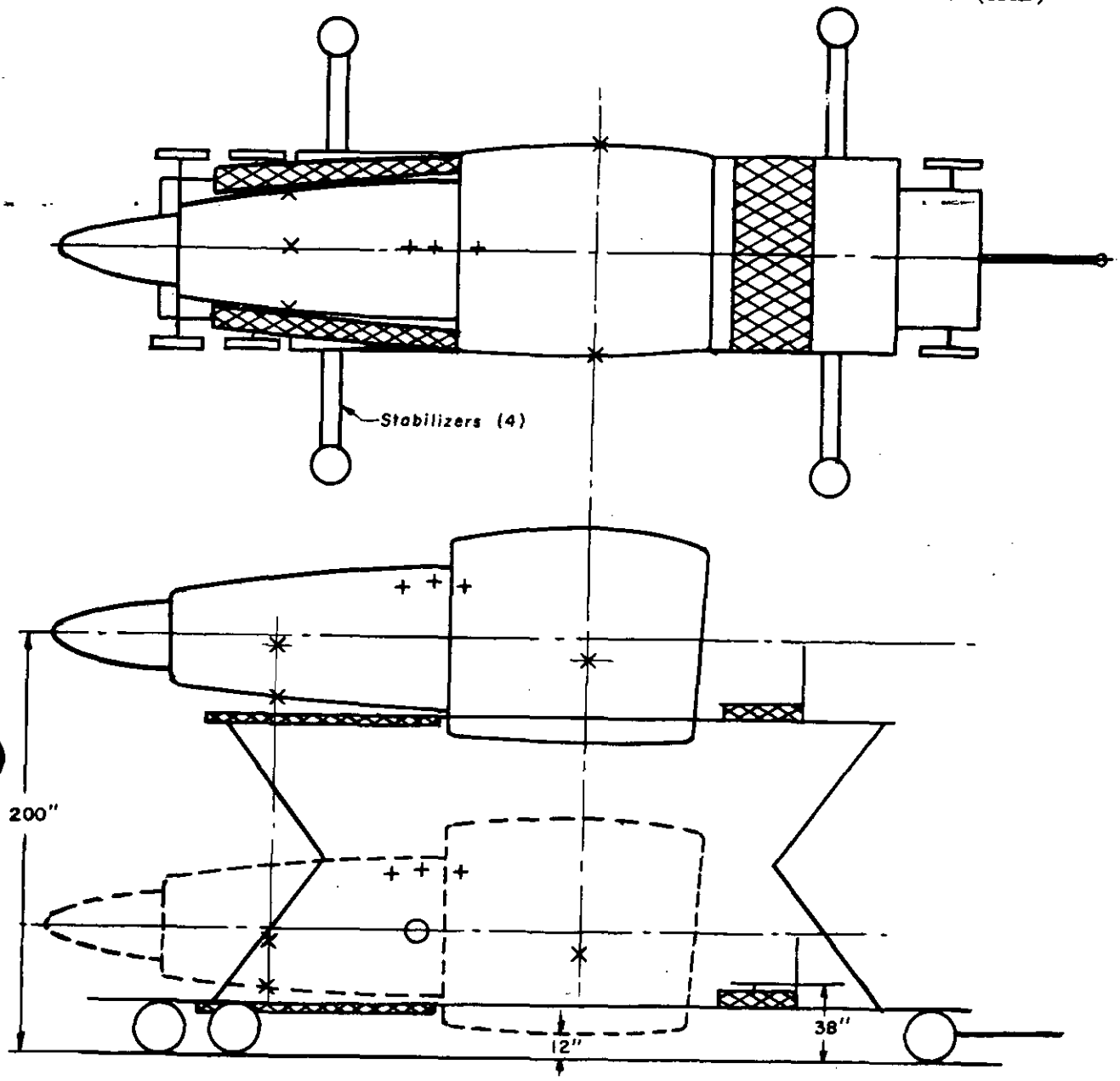
3.5.1 The trailer shall be designed and constructed to assure that all parts and components are so assembled as to be incapable of working loose during service. The trailer shall be designed to withstand the structural loads, accelerations, vibrations, and other conditions incident to mobile operation, shipping, storage, engine installation and removal, and service operations.

3.5.2 Interface requirements. The trailer, with and without the mounted engine, shall provide, but not be limited to, an interface capability with the following aerospace ground equipment items and aircraft:

a. Lockheed-Georgia Company interface control drawings 4BSAS00288, 4BSAS00290, 4BSAS00293, 4S30029, and 4S30040

b. Air transport, loading-tiedown and unloading in the C-5A, C-141A, and C-130 aircraft

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NOTE: X — ENGINE GROUND HANDLING MOUNTING POINTS
▩ — PLATFORM AREAS
+ — ENGINE FLIGHT MOUNTING POINTS

FIGURE 1. Typical Configuration of the Lift Trailer and Engine

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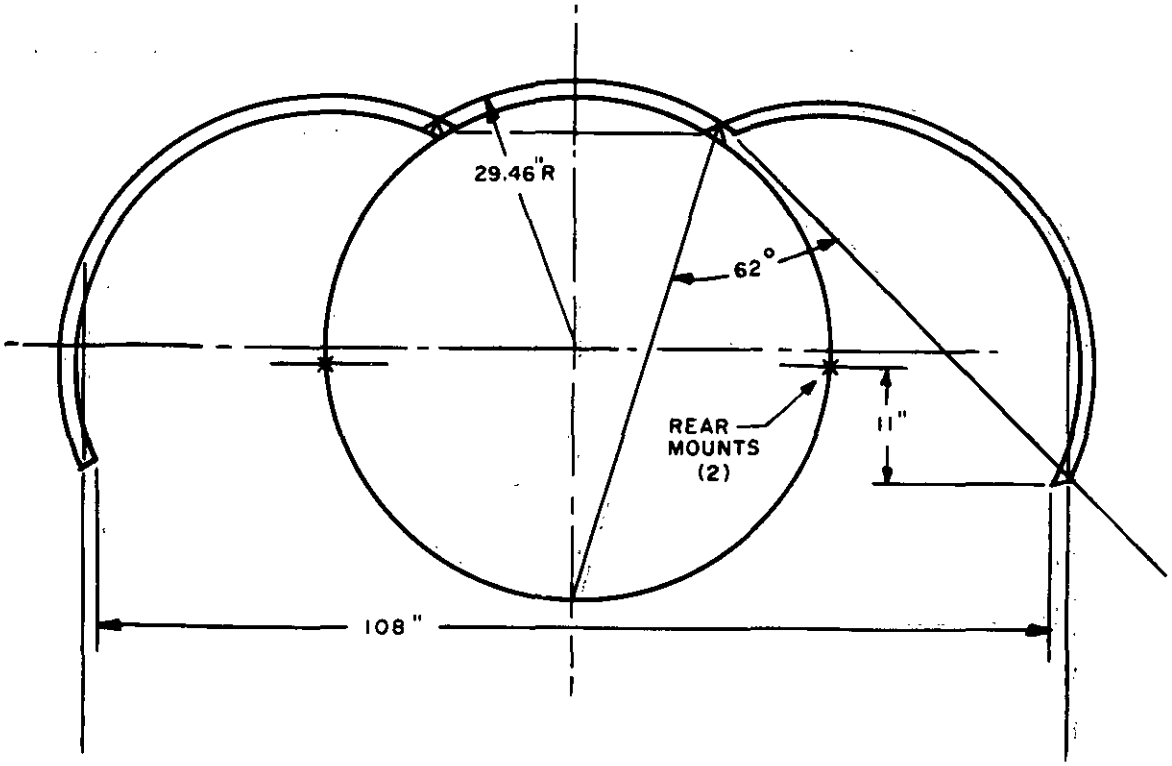
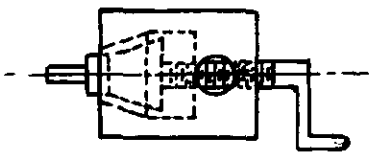
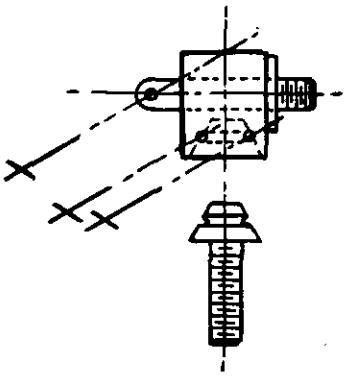


FIGURE 2. AIRCRAFT COWL DOOR OPEN CONFIGURATION

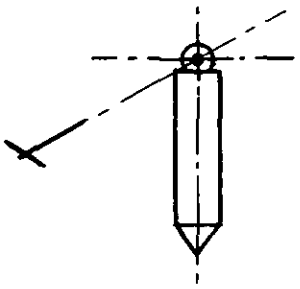
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REAR LATERAL MOUNT



FORWARD MOUNT



REAR LOWER MOUNT

FIGURE 3. Mounting Support Configuration

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c. Engine sling

d. Standard AF type tow vehicle

e. Trailer hoist and sling attach points

f. Rear-mounted pintle hook for towing trailer in train

g. Winch attach points for ramp loading (forward and aft)

h. A/M32A-60 generator set-power unit.

3.5.3 Structural capability. The trailer shall be designed to support a rated load of 12,000 pounds plus the weight of the trailer structure. The overall design load factors shall consider ground accelerations and vibrations in accordance with the requirements of MIL-M-8090 (see 6.2); air transport acceleration in accordance with the requirements of MIL-A-8421; and any other factors, which will govern the maximum loads to be encountered by the trailer during service. Structural members and components not governed by the preceding criteria shall be designed to include a factor of safety of three based on the material yield strength.

3.5.4 Reliability. The trailer shall have a minimum mean-time-between-failure (MTBF) of 500 hours with a minimum reliability of 0.998 for a 1-hour mission at a confidence factor of 0.90.

3.5.5 Maintainability. The trailer shall be designed and constructed in accordance with the following requirements and principles, wherever practicable. Any other provisions in this document that are more specific than the following shall be the governing provisions:

a. Parts and components shall be standardized to afford maximum practicable interchangeability. This shall include using military specification items, where applicable, and standardization within the equipment by using the same part numbered item in as many needed applications as practicable

b. The design shall facilitate visual inspection of parts expected to give a visual indication if and when failed

c. All maintenance shall require only general-purpose tools and test equipment commonly available commercially or in a maintenance shop appropriate to the equipment. Special tools, fixtures, or test equipment shall be subject to approval by the procuring activity

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d. Parts and components subject to replacement shall be positioned to give ample clearance for maintenance tasks and to be removed and replaced with minimum disturbance to other parts. Convenient fastenings, such as bolts and nuts, shall be used rather than permanent methods, such as rivets and welds, for items subject to replacement unless performance or reliability will be significantly degraded and unless the permanent fastening is common industry practice for similar applications

e. To the maximum practicable extent, ample clearances for maintenance actions and other features shall be provided to facilitate maintenance and servicing at low temperatures by personnel wearing heavy gloves or mittens and handicapped by bulky clothing and footgear

f. Provisions shall be included for complete removal and replacement of all parts that are subject to wear, deterioration, or damage. Provisions shall be included for lubrication of all parts subject to wear

g. Where improper installation of an item could cause malfunction of that item or the system in which it is installed, an unsymmetrical mounting means shall be provided. This mounting shall be so designed that the item can be installed only in its proper operating position.

3.5.6 Human engineering. At least the following recognized principles of human engineering shall be given consideration in the design of the trailer:

a. Human space limitations for operation and maintenance

b. The intellectual, physical, and psychomotor capabilities of the intended user

c. Visual and auditory perceptual requirements

d. Arrangements and readability of control and instrument displays

e. Safety factors covering minimization of potential human error in the operation and surveillance of the equipment

f. Comfort factors relative to efficiency of operation

g. Effects of climatic and mechanical service conditions on the ability of personnel to perform as required.

3.5.7 Air transportability. The trailer shall be air transportable in the C-5A, C-141A, and C-130 aircraft in an unloaded condition. The design of the trailer shall meet the requirements of MIL-A-8421, with the exceptions and additions as specified herein. Cargo compartment data is specified in Bulletin 518.

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3.5.7.1 Ramp loading. The ramp loading negotiation angle shall be 13.5°.

3.5.7.2 Load restraint. Load design restraints shall be as specified in MIL-A-8421, except the restraint capability shall be 8g in the forward and aft directions for emergency landing conditions.

3.5.7.3 Lifting and tiedown provisions. Lifting and tiedown eyes and attachments in accordance with MIL-STD-209 shall be provided on the trailer. The position and combined structural restraint shall be in accordance with MIL-A-8421.

3.5.8 Winch provisions. The trailer structure shall have provisions for winching from both ends.

3.5.9 Lubrication. The trailer shall be designed and constructed to operate satisfactorily when lubricated with the standard military lubricants specified in MIL-M-8090.

3.5.9.1 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541 and shall be located in accessible, protected positions. Extended fittings shall be provided to lubricate parts or assemblies that are not readily accessible for direct lubrication or which are likely to be overlooked because of inaccessibility.

3.5.10 Nonskid surfaces. Trailer structural members that personnel are likely to stand or climb on shall be coated with a nonskid type surface.

3.6 Performance

3.6.1 Environmental conditions. The trailer shall be capable of operating under and shall exhibit satisfactory resistance to the following:

a. Operating temperatures ranging from -65° to +120°F and storage temperatures from -80° to +160°F

b. Relative humidity of 95 to 100 percent

c. Exposure to salt-fog

d. Fungus growth as encountered in tropical climates

e. Dust as encountered in desert areas.

3.6.2 Mobility. The trailer shall withstand the stresses of mobile operation and provide shock load attenuation to the mounted engine as specified herein.

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3.6.3 Air transport. Air transportability pertaining to structural capability, acceleration loads, floor loading, ramp negotiation, and tiedown of an unloaded trailer shall be as specified herein.

3.6.4 Other transportation. Additional modes of transportation to be considered in the design of the trailer shall be over-the-road on a flatbed, truck-trailer, unloaded. This mode shall not be demonstrated.

3.6.5 Proof load. The trailer shall be capable of lifting a proof load of 24,000 pounds applied through the engine attachment fittings.

3.6.6 Side load. With the lift extended, the trailer shall be capable of withstanding a horizontal side load of 1,200 pounds.

3.6.7 Endurance. The trailer shall operate and prove satisfactory under the specified endurance test.

3.6.8 Life expectancy. The trailer shall be designed for a life expectancy of 10 years.

3.7 Chassis. The chassis shall be designed as a rigid, mobile structure that will serve as a mounting understructure for the power system, lift mechanism, and engine mounting frame. The chassis shall be equipped with heavy duty, steerable running gear and shall include power-operated stabilizer outriggers. The design shall provide for maximum stability and safety under all mobility, lift and load operations, specified herein.

3.7.1 Mobility. The design of the trailer shall incorporate the mobility requirements of MIL-M-8090 as specified for type II, group C equipment, with the exceptions and additions specified herein.

3.7.1.1 Towing speed. The towing speed of the trailer shall be 10 mph (maximum) loaded and 20 mph unloaded.

3.7.1.2 Parking brakes. Controls for operating the parking brakes shall be positioned on the forward end of the trailer. The brakes shall be designed to hold the loaded trailer, in either direction, on an incline of 13.5^{o}.

3.7.1.3 Service brakes. Air-over-hydraulic service brakes shall be provided on all wheels. The brakes shall be designed to provide safe control of the loaded trailer under all operating and mobility conditions. The brakes shall be designed to hold the loaded trailer, in either direction, on an incline of 13.5^{o}.

3.7.1.4 Wheels. The wheel configuration of the trailer shall give consideration to the use of single, super single, dual, or tandum arrangements.

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3.7.1.5 Tires. The tires shall be of a size and capacity adequate for handling and transporting the gross weight of the trailer under the specified operating conditions. The ground contact width of the trailer tires shall not exceed 96 inches.

3.7.1.6 Steering. Automotive-type steering mechanisms shall be provided on both ends of the trailer. The forward mechanism shall be towbar controlled and designed for the normal towing and ground mobility requirements specified herein. The rear steering shall be designed for manual steering and maneuvering and shall be controlled by a mechanically assisted mechanism. The forward towbar shall hinge into the vertical position and shall be equipped with stops to prevent contact with the trailer structure. The aft steering mechanism shall be equipped with a positive stop device to lock the mechanism in the center position during mobile operation.

3.7.2 Pintle hook. The trailer shall be provided with a rear-mounted pintle hook.

3.7.3 Stabilizers. Four power-operated stabilizer outriggers shall be provided. The stabilizers shall provide a capability to level the main chassis frame on a 3° sloping ramp surface in four directions, 90° apart, and shall laterally stabilize the entire unit under all lift and off-center load conditions. The stabilizers shall be equipped with base plates to distribute the load and shall have sufficient strength and extension to relieve the load from the pneumatic tires and suspension system during lifting operations. Each stabilizer shall be independently controlled and shall not extend beyond the overall width of the trailer when retracted. The lateral extension and dimensional configuration of the outriggers shall be compatible with the two prepositioned interface engine maintenance platforms (Drawing 4S30029).

3.7.3.1 Stabilizer controls. Controls for independently operating the stabilizers shall be mounted at the forward end of the trailer. The controls shall be arranged and actuated in relation to their function and shall provide maximum safety to the operator and equipment. Interlocks shall be provided to prevent operation of the lift mechanism unless the stabilizers have been extended and shall prevent operation of the stabilizers when the lift mechanism has been extended.

3.7.4 Level indicators. Bubble spirit levels shall be provided on the chassis of the trailer. The levels shall be utilized in the positioning of the stabilizers, and shall indicate leveling of the chassis frame within 1° of the true horizontal.

3.7.5 Shock-load-attenuation. The trailer shall be designed with an attenuation system to protect a mounted TF-39 engine from excessive accelerations and shock

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loads that may be experienced during ground mobility operations. The attenuation system shall prevent the trailer-mounted engine from receiving acceleration loads in excess of 2.5g (maximum) when subjected to the mobility tests specified in 4.6.3. The system shall incorporate such devices as shock absorbing mounts, wheel suspension, dampening devices, and combinations thereof.

3.8 Lift mechanism. The trailer shall be provided with a hydraulic power operated vertical lift mechanism, mounted within the chassis structural frame. In the retracted position, the lift structure shall provide a 12 ±1 inch clearance distance between the ground and lowest portion of the trailer-mounted engine fan cowling. In the extended position, the lift shall raise the aircraft engine a minimum of 200 inches above the ground, measured at the engine centerline. The forward end of the retracted lift and adjacent chassis structure shall not exceed 38 inches in height to allow the engine to be installed or removed by the interfacing aircraft power unit maintenance stand (Drawing 4S30040) or overhead facility hoist. The rate of lift/descent shall be 5 to 10 feet per minute. The lift shall not vary more than 1 inch from the true vertical, in any direction, from the retracted to the fully extended position and shall be equipped with a position-locking, fail-safe device that will hold the load in any extended position in the event of power system failure. The lift shall be designed to raise and lower a 12,000-pound load and shall be capable of lifting a proof load of 24,000 pounds while being subjected to a side load of 1,200 pounds. With the locking device disengaged and supporting a static proof load of 24,000 pounds, the settling rate of the lift mechanism shall not exceed 0.0625 inch per hour.

3.8.1 Lift hydraulic system. A complete self-contained hydraulic system shall be provided to power the lift mechanism. The hydraulic pump shall be driven by an electric motor operating from the power source specified in 3.10. The system shall be designed for use with hydraulic fluid conforming to MIL-H-5606, for continuous duty operation, and shall be furnished filled with the proper amount of fluid.

3.8.1.1 Hydraulic components. The hydraulic system shall contain, but need not be limited to, the following components: electric motor, hydraulic pump(s), hydraulic cylinders, valves, gages, coolers, reservoir, filter, controls, lines, and fittings. The reservoir shall be equipped with a fluid level indicator. Cylinder packing seals shall be of the O-ring type conforming to MIL-P-5516 with MS28774 V-type teflon back-up rings. Protective devices shall be incorporated to prevent overload in the motor and system components. Consideration shall be given to provide a capability to automatically unload pressure when the hydraulic system is in operation and the lift mechanism is not being activated.

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c. Yaw . . . . . $\pm 5^{0}$ in either direction about the engine nozzle exit centerline along the longitudinal centerline

d. Roll . . . . . $\pm 5^{0}$ in both directions about the engine centerline

e. Pitch . . . . . $+5^{0}$ with the nozzle in the vertical position about the lateral centerline

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aft engine mounting supports. The accuracy of each indicator shall be within ±10 percent of the actual load and shall be visible by an operator positioned on adjacent maintenance platforms or the trailer walkways, when the engine is being installed or removed from the aircraft pylon station.

3.10 Power system. All power operated equipment on the trailer shall operate directly or indirectly from an external power source of 115/208V ac, 400 Hz, three phase as supplied by an A/M32A-60 mobile generator set through an AN3430 cable assembly. The lift mechanism shall be powered by an electric motor driven hydraulic system. The stabilizers and mounting frame positioning mechanism shall be powered by electromechanical, hydromechanical drive systems, or combinations thereof. Power system equipment and components shall be accessible for servicing and maintenance.

3.10.1 Power input receptacle. The input power receptacle shall be an AN3114, with cover, that will mate with the ground power AN3430 cable assembly.

3.10.2 Auxiliary power outlet. A 115/208V ac, 400 Hz three-phase auxiliary outlet shall be furnished. The circuit shall be equipped with an MS3102R-22-22S receptacle and a 50-amp circuit breaker. The circuit shall be in parallel with the input power lines and shall tie-in between the AN3114 receptacle and the main power breaker.

3.10.3 Electrical system. All electrical components and wiring shall be weatherproof and conform to the requirements of Pamphlet 70 for class 1, division 1 locations. The power system shall be controlled by a main circuit breaker, with indicator light, and individual circuit breaker controls as required, located on the control panel. Motors shall be protected from reverse rotation in the event of incorrect phasing. The remote control pendant shall provide for an emergency shut-down switch to cut off all power to the electrical system in case of an emergency. The switch shall be isolated and designed to prevent inadvertent actuation.

3.10.4 Power system controls. The controls for operating the power system shall be grouped and conveniently located to provide maximum safety to the operation and equipment. The controls shall be located on a panel in relation to their function, easily accessible, and with clear-cut vision for the operator while the lift is being adjusted at any height position. All controls shall be readily accessible to an operator wearing gloves, and protected from accidental actuation. Controls and instruments shall be identified as to their function and so installed as to facilitate removal and servicing.

3.10.4.1 Control panel. A control panel shall be provided at the forward end of the trailer to contain the controls and instruments. The panel shall be mounted at a convenient height from the ground to facilitate observation and

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operation of the stabilizers and lift mechanism by an operator on the ground. The control panel shall be rigidly constructed and protected in a weatherproof enclosure, supported by a self-supporting framework.

3.11 Remote control pendant. The remote control pendant shall contain all of the positioning mechanism controls, the duplicated lift mechanism controls, and the emergency shutdown switch in a hand-held, weatherproof pendant control box. The pendant control box shall be connected to the trailer by an electrical cable of sufficient length to allow the control operator an access to all engine mounting areas. Provisions shall be incorporated to prevent inadvertent actuation of the controls. A storage compartment shall be provided on the engine mounting frame for the pendant and cable.

3.12 Personnel platforms. The upper portion of the lift mechanism shall be equipped with integral platforms that will provide operators with working access to the engine flight mounts and ground handling mounting points and provide a walkway across the forward end of trailer. The platforms and walkway shall be designed for 50 psf. The platforms and walkway shall be designed for use when the lift is in the elevated position and shall be compatible with the working surface and pull-outs of the two prepositioned interface engine maintenance platforms. The forward platform shall be equipped with a removable or collapsible guard rail. All platform areas shall be of a nonskid type surface.

3.13 Handbook compartment. A weatherproof compartment shall be provided for containing engine records and handbooks covering operation, service, and repair of the trailer. The compartment shall be a minimum size of 10 by 12 by 2 inches.

3.14 Storage compartment. A weatherproof storage compartment shall be provided on the trailer chassis for the storage of loose components. The compartment shall be a minimum size of 24 by 10 by 6 inches.

3.15 Plumb-bobs. Four type III, class B, 5-ounce plumb-bobs in accordance with GGG-P-501 shall be provided, except the cord length shall be 25 feet. The plumb-bobs shall have provisions for storage in the storage compartment.

3.16 Grounding cable. The reel shall conform to MIL-R-83232, except that in lieu of the type PCL electrical clip, an MS27610 connector shall be used. The reel shall be mounted on the forward or rear chassis structure.

3.17 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

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3.18 Electromagnetic interference. The trailer shall meet the requirements of MIL-STD-461 and MIL-STD-462 for class B equipment, radiated, method RE02, 14 KH_z to 1GH_z.

3.19 Dimensions. The overall dimensions of the trailer, with a mounted engine, shall be consistent with obtaining maximum stability of a towed vehicle and conforming to the mobility, air transportability, interface, and lift requirements specified herein.

3.20 Markings

3.20.1 Nameplates, instruction plates, data plates, and placards. All nameplates, instruction plates, data plates, and placards shall conform to MIL-P-514 and shall be of a size and shape consistent with the information required. Abbreviations shall be in accordance with MIL-STD-12. Mechanical symbols shall be in accordance with MIL-STD-17. All such plates shall be attached by metal screws or rivets.

3.20.2 Operational markings. The operational markings and stenciling shall be in accordance with the requirements of MIL-STD-808.

3.20.2.1 Tire pressure. The tire inflation pressure shall be stenciled on the trailer as near each wheel as practicable.

3.20.2.2 Towing speed. The towbar shall be marked as follows:

MAXIMUM TOWING SPEEDS: 10 MPH LOADED

20 MPH UNLOADED.

3.20.3 Transportation data plates. Transportation data plates shall be provided and shall be sized and marked in accordance with MIL-P-514.

3.20.3.1 Tiedown data plates. Data plates depicting tiedown instructions for ground and air transport shall be provided.

3.20.3.2 Marking of lifting and tiedown eyes. The location and capacity of each lifting and tiedown eye or attachment shall be clearly marked on the trailer.

3.20.4 Identification of controls and instruments. All gages, controls, and instruments used in the operation of the trailer shall be identified by securely attached nameplates of such composition that exposure to oil, dirt, light, et cetera, will not cause them to fade or become eradicated. Tags shall not be used.

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3.20.5 Operating instructions. Brief operating and precautionary instructions shall be permanently affixed on or near the control panel or other points as necessary. The instructions shall be clear, concise, and adequate to enable operation without damage to the system or injury to operating personnel. Precautionary instructions shall be so designed and located that the operator cannot fail to notice them before controls, valves, and switches are operated. Schematic drawings of the electrical and hydraulic systems shall be furnished.

3.20.5.1 Special operating instructions. Special operating instructions covering at least items such as engine installation and removal, the lift mechanism, positioning mechanism, safety locks, brakes, rear steering mechanism and power shall be permanently affixed near the operational controls.

3.20.6 Lubrication plate. A lubrication plate shall be provided directing attention to all lubrication fittings and shall specify the range and grade of lubricant required for operational temperatures. The plate shall be permanently attached to the trailer in an accessible and convenient location.

3.21 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.22 Workmanship. The trailer, including all parts and accessories, shall be constructed and finished in a workmanlike manner. Particular attention shall be given to neatness and thoroughness of marking of parts and assemblies, welding and brazing, painting, riveting, machine screw assemblies, and freedom of parts from burrs and sharp edges.

3.22.1 Riveting. Riveting operations shall be performed to insure that the rivets are tight and satisfactorily headed.

3.22.2 Cleaning. Prior to the application of the surface treatments, cleaning shall be accomplished as specified herein with materials and processes that have no deleterious effect on the metal and by means of a procedure that is satisfactory for the application of the surface treatment required.

3.22.3 Welding. Welding shall be accomplished by operators certified in accordance with MIL-T-5021, class A.

3.22.3.1 The trailer shall be thoroughly cleaned of loose, spattered, or excessive weld scale and flux; rust and metal chips; oil, grease, oxides, and other foreign matter after assembly. Burrs and sharp edges, as well as resin flux that may crumble, crack, et cetera, shall be removed.

3.22.3.2 Multiple-pass welds. When a weld is built up by two or more beads or passes, preceding beads shall be cleaned free of scale and flux by chipping or scraping and brushing.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the trailer shall be classified as follows:

a. Preproduction testing . . . . . See 4.4

b. Acceptance tests . . . . . See 4.5.

4.3 Test conditions

4.3.1 Standard test conditions. Unless otherwise specified herein, all tests shall be conducted under room conditions with the temperature at 25° ±10°C, the relative humidity at 90 percent or less, and the barometric pressure at 28 to 32 inches Hg.

4.3.2 Mockup engine. Deadweight loads equivalent to the configurational dimensions, 12,000 pound capacity, center of gravity position and mounting provisions simulating the engine, shall be provided and utilized for test purposes. The mockup engine shall be handled by sling and attachment fittings simulating those of the actual engine. A 24,000-pound load shall be required for the proof load test.

4.3.3 Instrumentation. The mockup engine and shock-attenuated portion of the trailer shall be instrumented with strategically positioned accelerometers and recording devices to determine the level of shock load accelerations and attenuation during the tests specified in 4.6.3.

4.3.4 Data. During the testing specified herein, sufficient data shall be recorded at proper intervals to insure compliance of the trailer with the performance requirements specified. Recorded data shall include the date, time, and ambient temperature at which the test is conducted as well as the data needed to accurately calculate the performance rates, efficiencies, et cetera, to be determined by the specified tests.

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4.3.5 Failure during test. Characteristics exhibited by the trailer during any of the tests specified herein that would cause the trailer to fail any of the requirements specified herein while in service shall constitute failure of the unit to pass the test and shall be the basis for rejection.

4.4 Preproduction testing (see 6.2)

4.4.1 Test sample. One trailer shall be subjected to the preproduction tests specified in 4.4.3.

4.4.2 Test report. Upon completion of the preproduction tests, a test report shall be prepared in accordance with MIL-STD-831 and three copies of the report furnished to the procuring activity.

4.4.2.1 Reliability and maintainability information. The following information shall be included as an appendix to the test report:

a. All failures, servicing, adjustments, maintenance, and irregular functioning shall be identified by accumulated operating time, cycles, miles, or position in the test procedure, as appropriate. Test conditions at the time of the events identified shall be recorded

b. A summary of the engineering analysis and of any tests conducted to determine assignable causes for any failure or irregular functioning

c. A summary of the engineering analysis leading to any corrections made to design, construction, quality control, or other procedures, or leading to any corrections to be made or proposed to be made to production items. The summary shall also include an analysis of the predicted effectiveness of such corrections

d. Clock time and man-hours required for each maintenance and servicing action taken during the tests. A brief description of the experience and qualifications of the personnel taking such actions

e. Test activity or contractor comments on item features or requirements that, if modified, should improve the item

f. Test activity or contractor comments on use or maintenance conditions to be avoided or cultivated to increase the reliability or useful life of the item

g. Any of the above information that is already included in the test report body need not be repeated in the information required by this paragraph, but clear reference to the location of the data shall be included.

4.4.3 Preproduction tests. The preproduction tests shall consist of all the applicable tests described under 4.6, conducted in the same order.

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4.4.3.1 Proposed test plan. A proposed test plan shall be submitted to the procuring activity for approval prior to initiation of testing. This plan shall describe the following:

a. Location of each test performed

b. Description of each test

c. Apparatus and facilities to be used

d. Test schedule showing starting and completion dates for each test.

4.5 Acceptance tests. The acceptance tests shall consist of the individual tests.

4.5.1 Individual tests. Each trailer shall be subjected to the following tests as described under 4.6:

a. Examination of product

b. Operational tests.

4.6 Test methods

4.6.1 Examination of product. The trailer shall be examined to determine conformance to the requirements of this specification with respect to dimensions, quality of workmanship, materials, design and finishes, markings, and visible defects, and the functional operation of all components, assemblies, accessories, and controls. Particular attention shall be given to the lift mechanism and its power supply system, the routing and protection of component parts, the positioning adjustments, and the overall structural stability of the trailer.

4.6.2 Operational tests. The trailer shall be operated under the following conditions to insure that the various components and capabilities comply with the requirements specified herein. During this performance demonstration, all components shall be operated according to the instructions provided with the trailer.

4.6.2.1 Functional. The trailer shall be functionally tested by placing the rated load into position on the trailer when the lift mechanism is in the retracted position. The load shall be mounted on the trailer with particular attention given to the compatibility of the trailer mountings and engine fixtures. With the stabilizers extended and locked into position, the trailer shall be operated through the entire range of lift and lowering. All controls, including manual, shall be checked for proper operation. The positive safety

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lock capability shall be demonstrated by producing a power failure to the system. With power off, the trailer shall be manually raised and lowered. During each cycle, when the load is at maximum height, the positioning mechanisms on the engine mount structure shall be both remotely and manually operated through their entire ranges. The load indicators shall be checked for accuracy. Any operational irregularity, instability, deformation, or failure shall be cause for rejection.

4.6.2.2 Leveling. Leveling of the trailer chassis shall be demonstrated in at least four different directions, 90° apart, on a 3° incline from the true horizontal.

4.6.3 Mobility. While supporting the 12,000-pound capacity load, the trailer shall be tested for type II, group C mobility in accordance with MIL-M-8090, with the following exceptions:

4.6.3.1 Towing distances. The towing distances shall be 100 miles over level paved highway type surface, 50 miles over graded gravel road, 50 miles over the type I test course, and 50 miles over the Belgian block course. The trailer shall be instrumented to verify that acceleration loads in excess of 2.5g are not being transmitted to the mock-up engine throughout the 250-mile towing test.

4.6.3.2 Towing speeds. Towing speeds for the paved and gravel surface roads shall be 10 ±1 mph and 5 ±1 mph for the type I test course and Belgian block course.

4.6.3.3 Rear steering. The manual operation and lockout capabilities of the rear steering mechanism shall be demonstrated.

4.6.3.4 Ramp. The ramp test shall be conducted on an unloaded trailer, except the incline shall be 13.5°. This test shall be conducted in the forward and reverse directions. The performance of the parking and service brakes shall be demonstrated with the 13.5° incline.

4.6.4 Air transportability. The unloaded trailer shall be tested for air transportability in accordance with MIL-A-8421.

4.6.5 Proof load. Hydraulic system adjustments shall be accomplished as required. The load indicators are not required to be operable for this test. After the stabilizers have been positioned, a proof load of 24,000 pounds shall be installed on the engine mounting supports and lifted to the maximum height. The load shall be held for 3 minutes. All positioning mechanism adjustments shall be operated through their full range of travel. While in the elevated position, a horizontal load of 1,200 pounds shall be applied to

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the load, midway between the supports at the engine centerline height. The horizontal load shall be retracted and the test repeated on the opposite side of the trailer. At the conclusion of the test, the lift shall be retracted. Any failure, permanent deformation, instability, or operational irregularity shall be cause for rejection.

4.6.5.1 Settling. During the tests specified in 4.6.5, the settling rate of the lift mechanism shall be measured. The settling rate shall not exceed 0.0625 inch per hour. The positive locking device shall not be used during the settling tests.

4.6.6 Endurance. Endurance tests shall be conducted on the trailer by operation of the power system, stabilizers, lift and positioning mechanisms. The test shall consist of 100 cycles of operation, as follows:

4.6.6.1 The trailer shall be subjected to 25 complete cycles of installing the 12,000-pound mock-up engine load on the trailer with the lift in the retracted position and lifting of the load to the maximum height. One cycle shall consist of installing the load on the engine supports, leveling of the chassis with the stabilizers, lifting of the load to the fully extended height, operation of all positioning mechanism functions and reverse operation.

4.6.6.2 The trailer shall be subjected to 25 complete cycles of installing the 12,000-pound mock-up engine on the unloaded trailer, when the load has been prepositioned approximately 200 inches above the ground level. One cycle shall consist of positioning the trailer directly under the load, stabilizing the chassis, elevating the lift, attaching the load to the trailer mounting supports by maneuvering of the positioning mechanism, and lowering the lift and mounted load to the fully retracted position. The cycle shall be completed after the load has been lifted and repositioned and the unloaded lift retracted.

4.6.6.3 The trailer shall be subjected to 50 cycles of lifting and lowering the 12,000-pound load after the load has been positioned on the mounting supports and the chassis frame stabilized. One cycle shall consist of lifting the load from the retracted position to the fully extended height and return to the retracted position.

4.6.6.4 Endurance test/inspection. Every fifth cycle during the endurance tests, the trailer will be inspected for defective component parts. During the endurance tests, the trailer shall be inspected for leakage. Fluid lines, fittings, pumps, and other components shall show no leakage. However, if such equipment can be tightened to stop leakage that may have occurred during the endurance test, such adjustments will be acceptable.

4.6.7 Electromagnetic interference. The trailer shall be tested as a complete unit for radiated electromagnetic interference in accordance with MIL-STD-462, class B, method REO2, 14 KH_Z to 1 GH_Z.

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4.6.8 Environmental tests. The loaded trailer shall be subjected to the following tests conducted in accordance with specified procedures of MIL-STD-810:

a. High temperature - Method 501, procedure I

b. Low temperature - Method 502, procedure I

c. Salt fog - Method 509, procedure I

d. Dust - Method 510, procedure I

e. Humidity - Method 507, procedure I

4.6.8.1 Fungus. Components of the hydraulic, electric, and brake systems that may be nutrients for fungi shall be tested in accordance with method 508, procedure II.

4.6.9 Reliability demonstration and test. Satisfactory completion of all tests specified herein shall be a demonstration of compliance with the quantitative reliability requirements of this specification.

4.6.10 Maintainability evaluation. The trailer shall be inspected and evaluated from the standpoint of ease of maintenance, servicing, and operation. Particular attention shall be given maintenance with a minimum number of common tools, servicing, and operation by personnel wearing heavy gloves, and provisions made to prevent accumulation of dirt, snow, ice, et cetera, that may hinder servicing and operation.

4.6.10.1 Preventive maintenance, servicing, and unit replacement. Maintenance tests shall be conducted to determine the adequacy and ease of performing preventive maintenance, servicing, and unit replacement. The frequency of servicing required for the various components shall be determined by study of existing supplier's recommendations and by servicing required during the preproduction tests. Interferences or obstructions to servicing shall be noted. Any tendency of the trailer to pick up mud, snow, and ice which interferes with servicing and operation shall be noted. Removal and installation of major components and assemblies to demonstrate ease of replacement shall be accomplished. Particular attention shall be given to time required, difficulties encountered, and the tools and skills required in conducting maintenance, servicing, and replacement.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing shall be in accordance with the instructions of the procuring activity.

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6. NOTES

6.1 Intended use. The ETU-92/E trailer is intended for transportation, installation, and removal operations for the C-5A aircraft TF-39 turbofan engine.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. That MIL-M-008090E(USAF), dated 18 April 1966, or later issue be used in conjunction with this specification

c. Location and conditions for preproduction testing, including proposed test plan (see 4.4)

6.3 Definition. For the purpose of this specification, the following definitions will apply:

6.3.1 EBU (Engine Build-up Unit). The EBU configuration of the TF-39 engine is defined as a bare engine with the fan cowling, thrust reverser, and tail cone installed.

6.4 Engine mounting supports. Lockheed-Georgia Company drawings 4S31318, 4P00002 MED (detail 679A), 4S31538, 4S31535, 4S31534, and 4S31532 will be provided for information purposes only. These drawings represent the design of engine mounting supports on a TF-39 engine transportation trailer, previously furnished to the Air Force. It is suggested the manufacturers compare the drawings with the engine mount requirements of this specification. If any questions arise, the engineering activity should be contacted for assistance.

Custodian:
Air Force - 11

Reviewer:
Air Force - 82

Preparing activity:
Air Force - 11

Project No. 1730-F145

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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